

Docker Layer Lab

Application is running successfully!

Environment: production

Python: 3.11

Port: 8000

Bước 3 — Tạo user-defined bridge network

```
docker network create app-net  
docker network ls
```

```
docker@docker:~$ docker network connect app-net c1  
docker@docker:~$ docker network connect app-net c2  
docker@docker:~$ docker exec c1 ping -c 2 c2  
PING c2 (172.20.0.5): 56 data bytes  
64 bytes from 172.20.0.5: seq=0 ttl=64 time=0.182 ms  
64 bytes from 172.20.0.5: seq=1 ttl=64 time=0.049 ms  
  
--- c2 ping statistics ---  
2 packets transmitted, 2 packets received, 0% packet loss  
round-trip min/avg/max = 0.049/0.115/0.182 ms
```

network lab

```
docker@docker:~$ docker run --rm -it --network lab-net alpine sh
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
55afalecc21d: Pull complete
56dceff11b33: Download complete
f5124fb579e2: Download complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
/ # ping lab-db
PING lab-db (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.101 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.059 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.044 ms
64 bytes from 172.18.0.2: seq=3 ttl=64 time=0.054 ms
64 bytes from 172.18.0.2: seq=4 ttl=64 time=0.056 ms
64 bytes from 172.18.0.2: seq=5 ttl=64 time=0.057 ms
^C
--- lab-db ping statistics ---
6 packets transmitted, 6 packets received, 0% packet loss
round-trip min/avg/max = 0.044/0.061/0.101 ms
```

Activate W
Go to Setting

docker DNS đã resolve

```
docker@docker:~$ docker exec -it demo-app bash
root@335168eb34a8:/app# getent hosts lab-db
172.18.0.2      lab-db
root@335168eb34a8:/app#
```

demo-app resolve được lab-db bằng container name, không cần biết IP PostgreSQL.

Docker Volume với PostgreSQL: tạo dữ liệu trong DB, xóa lab-db, tạo lại container mới và kiểm tra dữ liệu có còn hay không vào Postgre tạo database lab

```
2020-09-14 14:02:10.205 UTC [1] LOG:  database system is ready to accept connections
docker@docker:~$ docker exec -it lab-db psql -U labuser -d labdb
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

labdb=# CREATE TABLE users (
        id SERIAL PRIMARY KEY,
        name VARCHAR(100),
        email VARCHAR(100)
    );
CREATE TABLE
labdb=# INSERT INTO users (name, email)
VALUES
('Hung', 'hung@example.com'),
('Docker Lab', 'docker@example.com');
INSERT 0 2
labdb=# SELECT * FROM users;
 id |      name      |      email
----+-----+-----
  1 | Hung           | hung@example.com
  2 | Docker Lab     | docker@example.com
(2 rows)
```

Postgre đang dùng volume

```
docker@docker:~$ docker inspect lab-db | grep -A 10 Mounts
  "Mounts": [
    {
      "Type": "volume",
      "Name": "pg-data",
      "Source": "/var/lib/docker/volumes/pg-data/_data",
      "Destination": "/var/lib/postgresql/data",
      "Driver": "local",
      "Mode": "z",
      "RW": true,
      "Propagation": ""
    }
  ]
```

xóa container nhưng vẫn còn volume

```
docker@docker:~$ docker stop lab-db
lab-db
docker@docker:~$ docker rm lab-db
lab-db
docker@docker:~$ docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
335168eb34a8   docker-layer-lab:v1  "python app.py"         8 minutes ago   Up           0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
8f6bd80b6104   hello-world          "/hello"                50 minutes ago   Exited(0)    zen_bohr
bce891d4cd43   hello-world          "/hello"                About an hour ago   Exited(0)    hungry_raman

docker@docker:~$ docker volume ls
DRIVER          VOLUME NAME
local          07e340dc930f25d01aeac9713a80c818fa2ed368f1fd790aaeefe43f443f753c
local          pg-data
```

tạo lại container dùng chính volume cũ

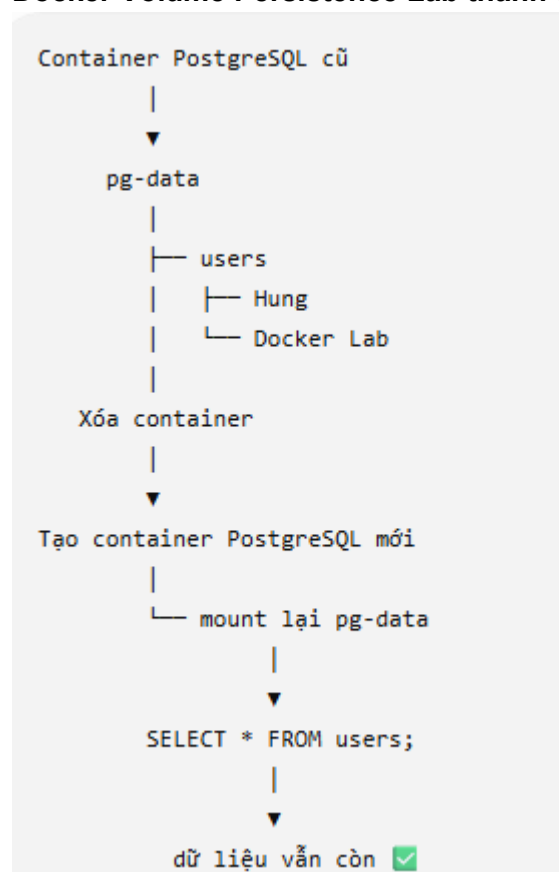
```
docker@docker:~$ docker volume ls
DRIVER          VOLUME NAME
local          07e340dc930f25d01aeac9713a80c818fa2ed368f1fd790aaeefe43f443f753c
local          pg-data
docker@docker:~$ docker run -d \
  --name lab-db \
  --network lab-net \
  -e POSTGRES_USER=labuser \
  -e POSTGRES_PASSWORD=labpass \
  -e POSTGRES_DB=labdb \
  -v pg-data:/var/lib/postgresql/data \
  postgres:16
bae0367caa8e214edd254da3b9f745a33a3df2bc8f203cc9199d23b0eca2d14a
docker@docker:~$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
bae0367caa8e   postgres:16          "docker-entrypoint.s...  4 s           Up           lab-db
335168eb34a8   docker-layer-lab:v1  "python app.py"         9 m           Up           0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
demo-app
```

kiểm tra data còn không

```
docker@docker:~$ docker exec -it lab-db psql -U labuser -d labdb
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

labdb=# SELECT * FROM users;
 id |      name      |      email
-----+-----+-----
  1 | Hung           | hung@example.com
  2 | Docker Lab     | docker@example.com
(2 rows)
```

Docker Volume Persistence Lab thành công !



Compose

```
docker@docker:~/docker-compose-lab$ ls -la
total 24
drwxrwxr-x 2 docker docker 4096 Sep 14 14:15 .
drwxr-x--- 8 docker docker 4096 Sep 14 14:15 ..
-rw-rw-r-- 1 docker docker  473 Sep 14 14:12 app.py
-rw-rw-r-- 1 docker docker  586 Sep 14 14:15 compose.yaml
-rw-rw-r-- 1 docker docker  222 Sep 14 14:15 Dockerfile
-rw-rw-r-- 1 docker docker   13 Sep 14 14:14 requirements.txt
docker@docker:~/docker-compose-lab$ docker compose config
```

test network thành công

```
[+] up 5/5
✓Image docker-compose-lab-web          Built
✓Network docker-compose-lab_lab-net    Created
✓Volume docker-compose-lab_pg-data     Created
✓Container lab-db                      Started
✓Container demo-app                    Started
docker@docker:~/docker-compose-lab$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE  CREATED      STATUS
demo-app            docker-compose-lab-web              "python app.py"                       web       6 seconds ago Up 6 seconds
0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
lab-db              postgres:16                          "docker-entrypoint.s..."           db        7 seconds ago Up 6 seconds
5432/tcp
docker@docker:~/docker-compose-lab$ curl http://localhost:8000
{"database_host":"db","environment":"production","hostname":"d4b5b1d774ee","message":"Hello from Docker Compose!"}
docker@docker:~/docker-compose-lab$ docker compose exec web bash
root@d4b5b1d774ee:/app# getent hosts db
172.19.0.2          db
root@d4b5b1d774ee:/app# |
```

test PostgreSQL

tạo dữ liệu

```
docker@docker:~/docker-compose-lab$ docker compose exec db psql -U labuser -d labdb
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

labdb=# CREATE TABLE compose_test (
        id SERIAL PRIMARY KEY,
        message VARCHAR(100)
    );

INSERT INTO compose_test (message)
VALUES ('Docker Compose persistence works');

SELECT * FROM compose_test;
CREATE TABLE
INSERT 0 1
 id |          message
----+-----
  1 | Docker Compose persistence works
(1 row)
```

restart lại compose

```
docker@docker:~/docker-compose-lab$ docker compose down
[+] down 3/3
✓Container demo-app Removed
✓Container lab-db Removed
✓Network docker-compose-lab_lab-net Removed
docker@docker:~/docker-compose-lab$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
docker@docker:~/docker-compose-lab$ docker compose up -d
[+] up 3/3
✓Network docker-compose-lab_lab-net Created
✓Container lab-db Started
✓Container demo-app Started
```

mở lại database posgre

```
docker@docker:~/docker-compose-lab$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED
demo-app docker-compose-lab-web "python app.py" web 4 seconds ago
lab-db postgres:16 "docker-entrypoint.s..." db 4 seconds ago
docker@docker:~/docker-compose-lab$ docker compose exec db psql -U labuser -d labdb
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

labdb=# SELECT * FROM compose_test;
 id | message
----+-----
  1 | Docker Compose persistence works
(1 row)

labdb=#
```

thành công!

```
docker compose down
↓
container bị xóa
network bị xóa
volume vẫn còn
↓
docker compose up -d
↓
container mới được tạo
↓
mount lại volume cũ
↓
data vẫn còn ✓
```

volume vẫn còn giữ !

```

docker@docker:~/docker-compose-lab$ docker volume ls
DRIVER      VOLUME NAME
local       07e340dc930f25d01aeac9713a80c818fa2ed368f1f
local       docker-compose-lab_pg-data
local       pg-data

```

! lưu ý nếu dùng docker compose down

lưu :stack + volume

! còn nếu dùng: docker compose down -v

có nghĩa xóa stack + xóa volume

```

docker@docker:~/docker-compose-lab$ docker compose up -v
unknown shorthand flag: 'v' in -v
docker@docker:~/docker-compose-lab$ docker compose up -d
[+] up 4/4
✓Network docker-compose-lab_lab-net Created
✓Volume docker-compose-lab_pg-data Created
✓Container lab-db Started
✓Container demo-app Started
docker@docker:~/docker-compose-lab$ docker compose exec db psql -U la
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

labdb=# SELECT * FROM compose_test;
ERROR:  relation "compose_test" does not exist
LINE 1: SELECT * FROM compose_test;
                        ^

labdb=# SELECT * FROM compose_test;
ERROR:  relation "compose_test" does not exist
LINE 1: SELECT * FROM compose_test;
                        ^

labdb=# |

```

docker tự phục hồi khi bị crash

```
09:20:16
START
|
| container chạy bình thường
| RestartCount=0
|
| sleep 30 giây
▼
09:20:46
CRASH
|
| exit 1
▼
Container dừng
|
| restart: unless-stopped
▼
Docker tự động restart
|
▼
09:20:46
START lần thứ 2

RestartCount=1 ✅
```

```

dk@dk:~/docker-compose-lab$ docker run -d \
  --name restart-test \
  --restart unless-stopped \
  alpine \
  sh -c 'echo "START $(date)"; sleep 30; echo "CRASH $(date)"; exit 1'
8691832c8783b80393c75f334f747c4f203dd2422ecf6787fee189d3383864ae
dk@dk:~/docker-compose-lab$ docker inspect \
  --format='Status={{.State.Status}} RestartCount={{.RestartCount}}' \
  restart-test
Status=running RestartCount=0
dk@dk:~/docker-compose-lab$ docker logs restart-test
START Tue Sep 15 09:20:16 UTC 2026
dk@dk:~/docker-compose-lab$ docker inspect \
  --format='Status={{.State.Status}} RestartCount={{.RestartCount}}' \
  restart-test
Status=running RestartCount=1
dk@dk:~/docker-compose-lab$ docker logs restart-test
START Tue Sep 15 09:20:16 UTC 2026
CRASH Tue Sep 15 09:20:46 UTC 2026
START Tue Sep 15 09:20:46 UTC 2026
dk@dk:~/docker-compose-lab$ |

```

giới hạn cpu

tạo container test

```

dk@dk:~$ docker run -d \
  --name cpu-test \
  --cpus="0.5" \
  alpine \
  sh -c 'while true; do ;; done'
6203ffc30090c007ba288dcfcb1428771d33217770124f5c5e6cbb5855ff50fa
dk@dk:~$ docker stats --no-stream cpu-test

```

troubleshoot giới hạn ram

tạo container ram limit

```

dk@dk:~$ docker run -d \
  --name memory-test \
  --memory="100m" \
  alpine \
  sleep 3600
6f8f468a930ac0ea943a3a0a978669693ef529117e3dd9c4de5caf1d75127089
dk@dk:~$ docker stats --no-stream memory-test

```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O
6f8f468a930a	memory-test	0.00%	468KiB / 100MiB	0.46%	876B /
126B	197kB / 0B	1			

Tạo container limit 100 MiB nhưng cố cấp phát 200 MB

Dùng Python image để cấp phát RAM một cách rõ ràng:

docker run -d \

--name memory-test \

```
--memory="100m" \
```

```
python:3.11-slim \
```

```
python -c 'x=bytearray(200*1024*1024); print("Allocated 200MB"); import time; time.sleep(3600)'
```

```
Memory-104857600 MemorySwap-2097152  
dk@dk:~$ sudo dmesg -T | tail -30
```

```
cope,task=python,pid=15699,uid=0  
[Wed Sep 16 02:01:17 2026] Memory cgroup out of memory: Killed process 15699  
(python) total-vm:217400kB, anon-rss:101232kB, file-rss:4864kB, shmem-rss:0  
kB, UID:0 pgtables:464kB oom_score_adj:0
```

cấu hình file compose.yaml giới hạn tài nguyên ram

Service **web** sẽ bắt đầu như sau:

web:

build: .

container_name: demo-app

restart: unless-stopped

cpus: 0.50

mem_limit: 256m

ports:

- "8000:8000"

Tương tự với db:

db:

image: postgres:16

container_name: lab-db

restart: unless-stopped

cpus: 0.75

mem_limit: 512m

```

dk@dk:~/docker-compose-lab$ docker inspect --format='Name={{.Name}} NanoCpus={{.HostConfig.NanoCpus}} Memory={{.HostConfig.Memory}}' demo-app
Name=/demo-app NanoCpus=500000000 Memory=268435456
dk@dk:~/docker-compose-lab$ docker inspect --format='Name={{.Name}} NanoCpus={{.HostConfig.NanoCpus}} Memory={{.HostConfig.Memory}}' lab-db
Name=/lab-db NanoCpus=750000000 Memory=536870912

```

backup

tạo đường dẫn backup riêng không nằm trong volume

```

dk@dk:~/docker-compose-lab$ mkdir -p ~/docker-backup-lab
dk@dk:~/docker-compose-lab$ ls -ld ~/docker-backup-lab
drwxrwxr-x 2 dk dk 4096 Sep 16 02:51 /home/dk/docker-backup-lab
dk@dk:~/docker-compose-lab$ docker exec lab-db \
pg_dump -U labuser -d labdb -Fc \
> ~/docker-backup-lab/labdb.dump
dk@dk:~/docker-compose-lab$ ls -lh ~/docker-backup-lab/labdb.dump
-rw-rw-r-- 1 dk dk 2.7K Sep 16 02:51 /home/dk/docker-backup-lab/labdb.dump

```

kiểm tra trước khi backup

```

dk@dk:~/docker-compose-lab$ docker exec -i lab-db \
pg_restore -l \
< ~/docker-backup-lab/labdb.dump
;
; Archive created at 2026-09-16 02:51:08 UTC
; dbname: labdb
; TOC Entries: 11
; Compression: gzip
; Dump Version: 1.15-0
; Format: CUSTOM
; Integer: 4 bytes
; Offset: 8 bytes
; Dumped from database version: 16.15 (Debian 16.15-1.pgdg13+2)
; Dumped by pg_dump version: 16.15 (Debian 16.15-1.pgdg13+2)
;
;
; Selected TOC Entries:
;
216; 1259 24578 TABLE public users labuser
215; 1259 24577 SEQUENCE public users_id_seq labuser
3421; 0 0 SEQUENCE OWNED BY public users_id_seq labuser
3267; 2604 24581 DEFAULT public users id labuser
3414; 0 24578 TABLE DATA public users labuser
3422; 0 0 SEQUENCE SET public users_id_seq labuser
3269; 2606 24583 CONSTRAINT public users users_pkey labuser

```

xóa dữ liệu

```

dk@dk:~/docker-compose-lab$ docker exec lab-db \
  psql -U labuser -d labdb \
  -c "DELETE FROM users;"
DELETE 3
dk@dk:~/docker-compose-lab$ docker exec lab-db \
  psql -U labuser -d labdb \
  -c "SELECT * FROM users ORDER BY id;"
id | name | email
----+-----+-----
(0 rows)

dk@dk:~/docker-compose-lab$ curl -s http://localhost:8000/users
[]

```

restore dữ liệu

```

dk@dk:~/docker-compose-lab$ docker exec -i lab-db \
  pg_restore \
  -U labuser \
  -d labdb \
  --clean \
  --if-exists \
  < ~/docker-backup-lab/labdb.dump
dk@dk:~/docker-compose-lab$ echo $?
0

```

kiểm tra

```

dk@dk:~/docker-compose-lab$ docker exec lab-db \
  psql -U labuser -d labdb \
  -c "SELECT * FROM users ORDER BY id;"
id | name | email
----+-----+-----
1 | Hung | hung@example.com
2 | Docker | docker@example.com
3 | Backup | backup@example.com
(3 rows)

dk@dk:~/docker-compose-lab$ curl -s http://localhost:8000/users
[{"email":"hung@example.com","id":1,"name":"Hung"}, {"email":"docker@example.com","id":2,"name":"Docker"}, {"email":"backup@example.com","id":3,"name":"Backup"}]
dk@dk:~/docker-compose-lab$ curl -s -X POST http://localhost:8000/users \
  -H "Content-Type: application/json" \
  -d '{"name":"AfterRestore","email":"afterrestore@example.com"}'
{"email":"afterrestore@example.com","id":4,"name":"AfterRestore"}

```

Giả lập mất Container + Volume

xóa container + volume

```
dk@dk:~/docker-compose-lab$ docker compose down -v
[+] down 4/4
✓Container demo-app          Removed
✓Container lab-db            Removed
✓Volume docker-compose-lab_pg-data Removed
✓Network docker-compose-lab_lab-net Removed
dk@dk:~/docker-compose-lab$ |
```

kiểm tra

```
dk@dk:~/docker-compose-lab$ docker compose ps -a
NAME          IMAGE          COMMAND          SERVICE    CREATED   STATUS    PORTS
dk@dk:~/docker-compose-lab$ docker volume ls | grep docker-compose-lab_pg-da
ta
dk@dk:~/docker-compose-lab$ ls -lh ~/docker-backup-lab/labdb.dump
-rw-rw-r-- 1 dk dk 2.7K Sep 16 02:51 /home/dk/docker-backup-lab/labdb.dump
dk@dk:~/docker-compose-lab$ |
```

dựng lại compose

```
dk@dk:~/docker-compose-lab$ docker compose up -d
[+] up 4/4
✓Network docker-compose-lab_lab-net Created      0.0s
✓Volume docker-compose-lab_pg-data Created      0.0s
✓Container lab-db             Healthy     5.7s
✓Container demo-app           Started     5.8s
```

toàn bộ dữ liệu mới

```
dk@dk:~/docker-compose-lab$ docker volume ls | grep docker-compose-lab_pg-da
ta
local        docker-compose-lab_pg-data
dk@dk:~/docker-compose-lab$ docker exec lab-db \
    psql -U labuser -d labdb \
    -c "\dt"
Did not find any relations.
dk@dk:~/docker-compose-lab$ curl -i http://localhost:8000/users
HTTP/1.1 500 INTERNAL SERVER ERROR
Server: Werkzeug/3.1.8 Python/3.11.16
Date: Wed, 16 Sep 2026 03:08:45 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 265
Connection: close

<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your
request. Either the server is overloaded or there is an error in the applica
tion.</p>
```

restore labdb.dump vào PostgreSQL

dữ liệu trả về đúng ban đầu

```

dk@dk:~/docker-compose-lab$ docker exec -i lab-db \
pg_restore \
-U labuser \
-d labdb \
< ~/docker-backup-lab/labdb.dump
dk@dk:~/docker-compose-lab$ echo $?
0
dk@dk:~/docker-compose-lab$ docker exec lab-db \
psql -U labuser -d labdb \
-c "\dt"
      List of relations
Schema | Name  | Type  | Owner
-----+-----+-----+-----
public | users | table | labuser
(1 row)

dk@dk:~/docker-compose-lab$ docker exec lab-db \
psql -U labuser -d labdb \
-c "SELECT * FROM users ORDER BY id;"
 id | name  | email
----+-----+-----
  1 | Hung  | hung@example.com
  2 | Docker| docker@example.com
  3 | Backup| backup@example.com
(3 rows)

```

kiểm tra dữ liệu sau restore

```

dk@dk:~/docker-compose-lab$ curl -i http://localhost:8000/users
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.11.16
Date: Wed, 16 Sep 2026 03:11:54 GMT
Content-Type: application/json
Content-Length: 160
Connection: close

[{"email":"hung@example.com","id":1,"name":"Hung"}, {"email":"docker@example.com","id":2,"name":"Docker"}, {"email":"backup@example.com","id":3,"name":"Backup"}]
dk@dk:~/docker-compose-lab$ curl -s -X POST http://localhost:8000/users \
-H "Content-Type: application/json" \
-d '{"name":"DisasterRecovery","email":"dr@example.com"}'
{"email":"dr@example.com","id":4,"name":"DisasterRecovery"}

```

NginX

```

dk@dk:~/docker-compose-lab$ mkdir -p nginx
dk@dk:~/docker-compose-lab$ vim nginx/default.conf
dk@dk:~/docker-compose-lab$ cat nginx/default.conf
server {
    listen 80;
    server_name _;

    location / {
        proxy_pass http://web:8000;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

```

cấu hình file compose.yaml

```

nginx:
  container_name: nginx-proxy
  depends_on:
    web:
      condition: service_healthy
      required: true
  image: nginx:alpine
  networks:
    lab-net: null
  ports:
    - mode: ingress
      target: 80
      published: "80"
      protocol: tcp
  restart: unless-stopped
  volumes:
    - type: bind
      source: /home/dk/docker-compose-lab/nginx/default.conf
      target: /etc/nginx/conf.d/default.conf
      read_only: true
      bind: {}

```

mở compose

```

dk@dk:~/docker-compose-lab$ docker compose up -d
[+] up 13/13
✔Image nginx:alpine      Pulled          7.0s
✔Container demo-app      Healthy       1.1s
✔Container lab-db        Healthy       0.6s
✔Container nginx-proxy    Started       1.3s

```

check compose nginx và web

```

dk@dk:~/docker-compose-lab$ curl -i http://localhost/
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Wed, 16 Sep 2026 03:40:27 GMT
Content-Type: application/json
Content-Length: 115
Connection: keep-alive

{"database_host":"db","environment":"production","hostname":"3a3579399419","message":"Hello from Docker Compose!"}
dk@dk:~/docker-compose-lab$ curl -i http://localhost/users
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Wed, 16 Sep 2026 03:41:01 GMT
Content-Type: application/json
Content-Length: 220
Connection: keep-alive

[{"email":"hung@example.com","id":1,"name":"Hung"}, {"email":"docker@example.com","id":2,"name":"Docker"}, {"email":"backup@example.com","id":3,"name":"Backup"}, {"email":"dr@example.com","id":4,"name":"DisasterRecovery"}]

```

check log

```

dk@dk:~/docker-compose-lab$ docker logs --tail 10 nginx-proxy
2026/09/16 03:38:37 [notice] 1#1: using the "epoll" event method
2026/09/16 03:38:37 [notice] 1#1: nginx/1.31.6
2026/09/16 03:38:37 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/09/16 03:38:37 [notice] 1#1: OS: Linux 6.8.0-139-generic
2026/09/16 03:38:37 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/09/16 03:38:37 [notice] 1#1: start worker processes
2026/09/16 03:38:37 [notice] 1#1: start worker process 20
2026/09/16 03:38:37 [notice] 1#1: start worker process 21
172.19.0.1 - - [16/Sep/2026:03:40:27 +0000] "GET / HTTP/1.1" 200 115 "-" "curl/8.5.0" "-"
172.19.0.1 - - [16/Sep/2026:03:41:01 +0000] "GET /users HTTP/1.1" 200 220 "-" "curl/8.5.0" "-"
dk@dk:~/docker-compose-lab$

```

Đóng port backend 8000

Trong service `web`, tìm:

```

YAML

ports:
  - "8000:8000"

```

Xóa hai dòng đó.

khởi động lại compose

```

name: docker-compose-lab_pg_data
dk@dk:~/docker-compose-lab$ docker compose up -d
[+] up 3/3
✓Container lab-db           Healthy      10.7s
✓Container nginx-proxy      Running      0.0s
✓Container demo-app         Healthy      15.8s
dk@dk:~/docker-compose-lab$ docker compose ps
NAME                IMAGE                                COMMAND                  SERVICE    CR
demo-app            docker-compose-lab-web             "python app.py"         web        20
seconds ago        Up 9 seconds (healthy)          8000/tcp
lab-db              postgres:16                         "docker-entrypoint.s..." db         38
minutes ago        Up 38 minutes (healthy)        5432/tcp
nginx-proxy         nginx:alpine                       "/docker-entrypoint...." nginx      5
minutes ago        Up 5 minutes                    0.0.0.0:80->80/tcp, [::]:80->80/tcp

```

qua nginx thành công

```

dk@dk:~/docker-compose-lab$ curl -i http://localhost/users
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Wed, 16 Sep 2026 03:44:55 GMT
Content-Type: application/json
Content-Length: 220
Connection: keep-alive

[{"email":"hung@example.com","id":1,"name":"Hung"}, {"email":"docker@example.com","id":2,"name":"Docker"}, {"email":"backup@example.com","id":3,"name":"Backup"}, {"email":"dr@example.com","id":4,"name":"DisasterRecovery"}]

```

bypass nginx thất bại

```

dk@dk:~/docker-compose-lab$ curl -i --connect-timeout 3 http://localhost:8000/users
curl: (7) Failed to connect to localhost port 8000 after 0 ms: Couldn't connect to server

```

nginx vẫn truy cập backend bình thường

```

dk@dk:~/docker-compose-lab$ docker exec nginx-proxy wget -qO- http://web:8000/health
{"status":"OK"}

```

Host → :80 → Nginx → demo-app PASS ✓

Host → :8000 → demo-app BLOCKED ✓

Nginx → web:8000 → demo-app PASS ✓

Nginx là cổng vào duy nhất (entry point) của ứng dụng. Client không cần biết Flask đang chạy ở đâu hay port nào; chỉ cần truy cập Nginx qua `:80`.

Cụ thể, nó mang lại 4 lợi ích chính:

- Ẩn backend: `demo-app:8000` không publish ra host; client không bypass Nginx được.
- Chuyển tiếp request: ví dụ `GET http://server/users` → Nginx → `http://web:8000/users`.
- Tập trung kiểm soát traffic: sau này có thể thêm HTTPS/TLS, logging, rate limit, security headers ngay tại Nginx.
- Hỗ trợ mở rộng: có thể route/load balance tới nhiều backend mà client vẫn chỉ truy cập một địa chỉ.

Có thể nhớ bằng một câu:

Nginx Reverse Proxy đứng trước backend, nhận request từ client và thay mặt client chuyển request tới đúng application phía sau.

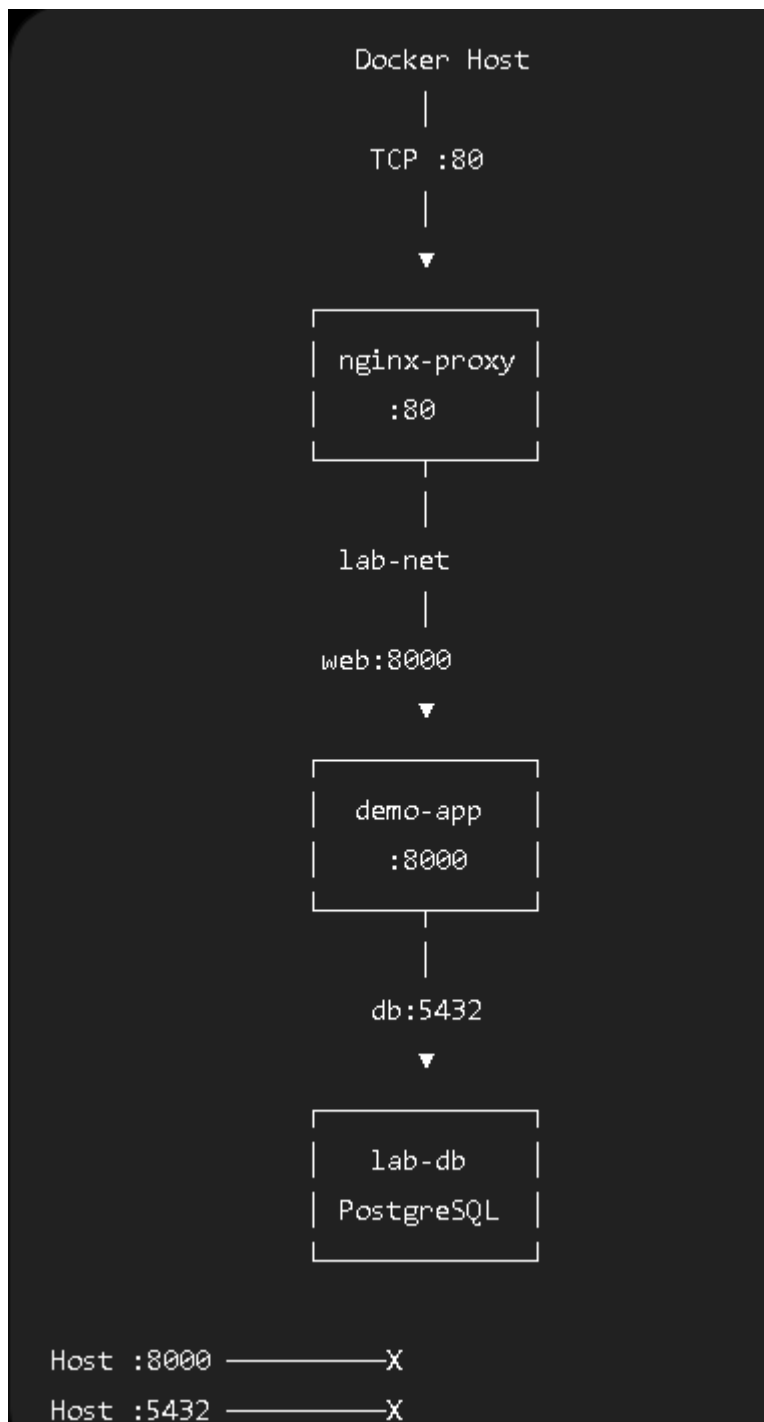
Nginx là cổng vào duy nhất (entry point) của ứng dụng. Client không cần biết Flask đang chạy ở đâu hay port nào; chỉ cần truy cập Nginx qua `:80`.

Cụ thể, nó mang lại 4 lợi ích chính:

- Ẩn backend: `demo-app:8000` không publish ra host; client không bypass Nginx được.
- Chuyển tiếp request: ví dụ `GET http://server/users` → Nginx → `http://web:8000/users`.
- Tập trung kiểm soát traffic: sau này có thể thêm HTTPS/TLS, logging, rate limit, security headers ngay tại Nginx.
- Hỗ trợ mở rộng: có thể route/load balance tới nhiều backend mà client vẫn chỉ truy cập một địa chỉ.

Có thể nhớ bằng một câu:

Nginx Reverse Proxy đứng trước backend, nhận request từ client và thay mặt client chuyển request tới đúng application phía sau.



Tách thành 2 Docker Network

```
dk@dk:~/docker-compose-lab$ docker network ls
NETWORK ID          NAME                                     DRIVER          SCOPE
8a7e5a75aadd        bridge                                bridge          local
293d4e2cdd9c        docker-compose-lab_backend-net        bridge          local
8168d08c69ff        docker-compose-lab_frontend-net       bridge          local
ecc5064a6e4d        docker-compose-lab_lab-net            bridge          local
2ad4918903cb        host                                  host            local
ad4f381062a4        lab-net                               bridge          local
c83fcca1ed43        none                                  null            local
```

cấu hình

service nginx
networks:
- frontend-net

service web
networks:
- frontend-net
- backend-net

Service db
networks:
- backend-net

test network

nginx và web cùng frontend-net

nginx và db không cùng network

demo-app — backend-net — lab-db

```
dk@dk:~/docker-compose-lab$ docker exec nginx-proxy getent hosts web
172.21.0.2      web  web
dk@dk:~/docker-compose-lab$ docker exec nginx-proxy getent hosts db
dk@dk:~/docker-compose-lab$ echo $?
2
dk@dk:~/docker-compose-lab$ docker exec demo-app getent hosts db
172.20.0.2      db
```

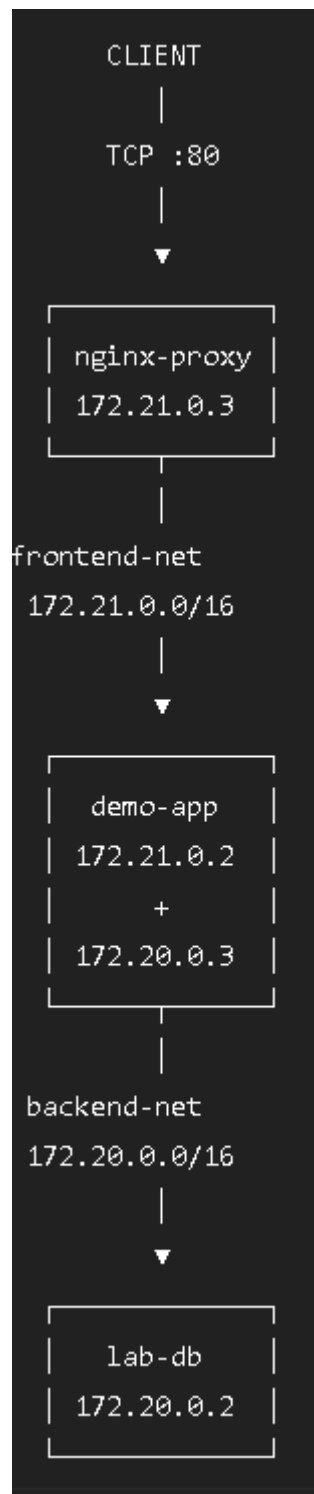
demo-app

```
|— 172.21.0.2 ← frontend-net
|— 172.20.0.3 ← backend-net
```

```
dk@dk:~/docker-compose-lab$ docker network inspect docker-compose-lab_frontend-net \
--format '{{range .Containers}}{{.Name}} -> {{.IPv4Address}}{{println}}{{end}}'
demo-app -> 172.21.0.2/16
nginx-proxy -> 172.21.0.3/16

dk@dk:~/docker-compose-lab$ docker network inspect docker-compose-lab_backend-net \
--format '{{range .Containers}}{{.Name}} -> {{.IPv4Address}}{{println}}{{end}}'
demo-app -> 172.20.0.3/16
lab-db -> 172.20.0.2/16

dk@dk:~/docker-compose-lab$ |
```



Docker Security / Hardening

```
dk@dk:~/docker-compose-lab$ docker compose config | grep -A2 security_opt
security_opt:
  - no-new-privileges:true
networks:
```

Drop capabilities của demo-app

cap_drop:

- ALL

security_opt:

- no-new-privileges:true

```
dk@dk:~/docker-compose-lab$ docker compose up -d web
[+] up 2/2
  ✓ Container lab-db    Healthy          10.7s
  ✓ Container demo-app  Started         10.9s
dk@dk:~/docker-compose-lab$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE  CR
demo-app            docker-compose-lab-web             "python app.py"                      web       35
seconds ago        Up 23 seconds (healthy)             8000/tcp
lab-db              postgres:16                         "docker-entrypoint.s..."           db        54
minutes ago        Up 53 minutes (healthy)             5432/tcp
nginx-proxy         nginx:alpine                       "/docker-entrypoint...."           nginx     54
minutes ago        Up 53 minutes                       0.0.0.0:80->80/tcp, [::]:80->80/tcp
dk@dk:~/docker-compose-lab$ docker exec demo-app sh -c \
'grep -E "(CapInh|CapPrm|CapEff|CapBnd|CapAmb):" /proc/1/status'
CapInh: 0000000000000000
CapPrm: 0000000000000000
CapEff: 0000000000000000
CapBnd: 0000000000000000
CapAmb: 0000000000000000
dk@dk:~/docker-compose-lab$
```

Hardening hiện tại của demo-app

```
demo-app
├─ UID/GID 10001 appuser          ✓ Non-root
├─ NoNewPrivs = 1                 ✓ Chặn tăng privilege
├─ CapInh = 0                     ✓
├─ CapPrm = 0                     ✓
├─ CapEff = 0                     ✓ Không Linux capabilities
├─ CapBnd = 0                     ✓
├─ CapAmb = 0                     ✓
├─ Health = healthy              ✓
└─ /users = HTTP 200             ✓ Application vẫn hoạt động
```

Read-only

```

dk@dk:~/docker-compose-lab$ docker compose config | grep -E -A2 'read_only|c
ap_drop|security_opt'
    read_only: true
    bind: {}
web:
--
    cap_drop:
    - ALL
    cpus: 0.5
--
    read_only: true
    restart: unless-stopped
    security_opt:
    - no-new-privileges:true
networks:

```

check file sau khi up compose

```

dk@dk:~/docker-compose-lab$ docker compose up -d web
[+] up 2/2
✓Container lab-db    Healthy      10.7s
✓Container demo-app  Started     10.9s
dk@dk:~/docker-compose-lab$ ^C
dk@dk:~/docker-compose-lab$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE  CR
demo-app            docker-compose-lab-web             "python app.py"                      web      19
seconds ago        Up 8 seconds (healthy)              8000/tcp
lab-db              postgres:16                         "docker-entrypoint.s..."          db       Ab
out an hour ago   Up About an hour (healthy)          5432/tcp
nginx-proxy         nginx:alpine                       "/docker-entrypoint...."          nginx    Ab
out an hour ago   Up About an hour                    0.0.0.0:80->80/tcp, [::]:80->
80/tcp
dk@dk:~/docker-compose-lab$ docker exec demo-app sh -c \
'echo "security-test" > /tmp/security-test-2.txt'
sh: 1: cannot create /tmp/security-test-2.txt: Read-only file system

```

web vẫn up

```

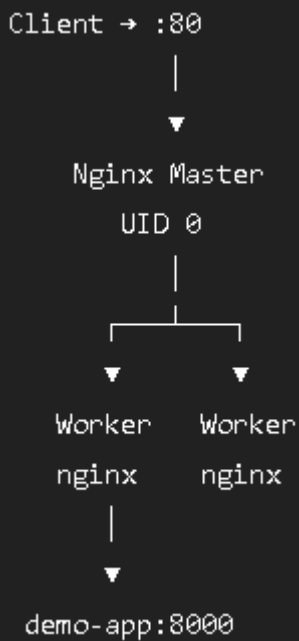
dk@dk:~/docker-compose-lab$ docker exec demo-app sh -c \
'echo "# hacked" >> /app/app.py'
sh: 1: cannot create /app/app.py: Read-only file system
dk@dk:~/docker-compose-lab$ curl -i http://localhost/users
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Wed, 16 Sep 2026 07:30:15 GMT
Content-Type: application/json
Content-Length: 220
Connection: keep-alive

[{"email":"hung@example.com","id":1,"name":"Hung"}, {"email":"docker@example.
com", "id":2, "name":"Docker"}, {"email":"backup@example.com", "id":3, "name":"Ba
ckup"}, {"email":"dr@example.com", "id":4, "name":"DisasterRecovery"}]

```

```
dk@dk:~/docker-compose-lab$ docker inspect demo-app --format \
'User={{.Config.User}}
ReadonlyRootfs={{.HostConfig.ReadonlyRootfs}}
SecurityOpt={{json .HostConfig.SecurityOpt}}
CapDrop={{json .HostConfig.CapDrop}}'
User=appuser
ReadonlyRootfs=true
SecurityOpt=["no-new-privileges:true"]
CapDrop=["ALL"]
```

```
dk@dk:~/docker-compose-lab$ docker exec nginx-proxy ps -o pid,user,group,com
m
PID    USER    GROUP    COMMAND
1      root     root     nginx
21     nginx   nginx   nginx
22     nginx   nginx   nginx
65     root     root     ps
```



PostgreSQL baseline

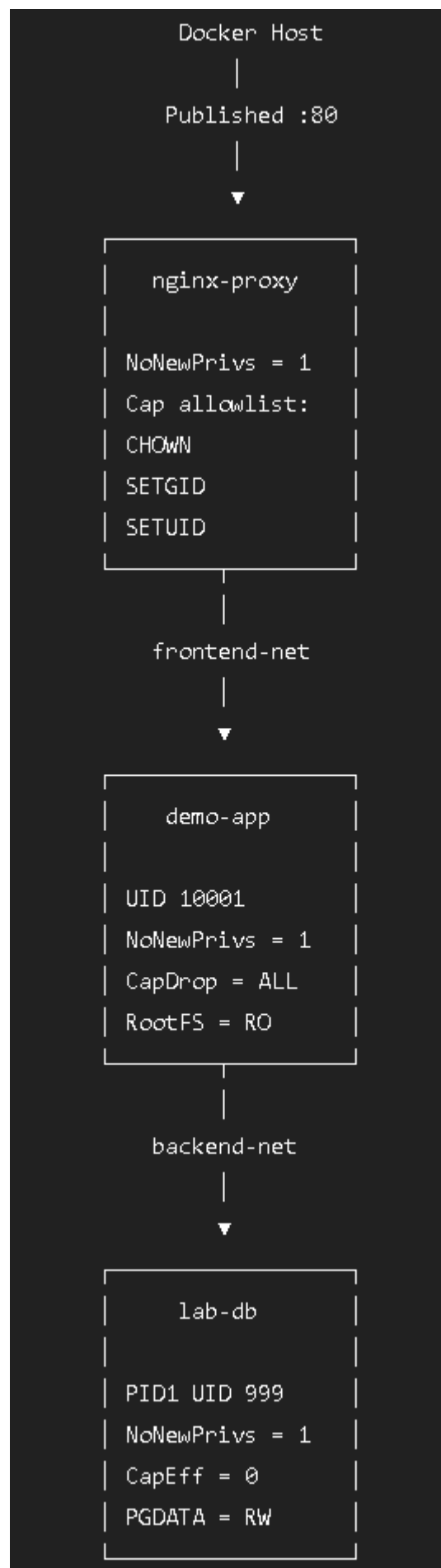
```

dk@dk:~/docker-compose-lab$ docker exec lab-db sh -c \
'grep -E "^(Name|Uid|Gid|NoNewPrivs|CapInh|CapPrm|CapEff|CapBnd|CapAmb):" /p
roc/1/status'
Name:      postgres
Uid:       999      999      999      999
Gid:       999      999      999      999
CapInh:    0000000000000000
CapPrm:    0000000000000000
CapEff:    0000000000000000
CapBnd:    00000000a80425fb
CapAmb:    0000000000000000
NoNewPrivs: 0
dk@dk:~/docker-compose-lab$ docker inspect lab-db --format \
'User={{json .Config.User}}
ReadonlyRootfs={{.HostConfig.ReadonlyRootfs}}
SecurityOpt={{json .HostConfig.SecurityOpt}}
CapDrop={{json .HostConfig.CapDrop}}
CapAdd={{json .HostConfig.CapAdd}}'
User=""
ReadonlyRootfs=false
SecurityOpt=null
CapDrop=null
CapAdd=null
dk@dk:~/docker-compose-lab$ docker exec lab-db sh -c \
'id && echo "PGDATA=$PGDATA" && ls -ld "$PGDATA"'
uid=0(root) gid=0(root) groups=0(root)
PGDATA=/var/lib/postgresql/data
drwx----- 19 postgres postgres 4096 Sep 16 06:22 /var/lib/postgresql/data
dk@dk:~/docker-compose-lab$

```

PostgreSQL

- |— Không publish 5432 ra Host 
- |— Chỉ nằm backend-net 
- |— Nginx không truy cập trực tiếp DB 
- |— PostgreSQL PID1 chạy UID 999 
- |— PostgreSQL CapEff = 0 
- |— no-new-privileges = true 
- |— Resource limit CPU/RAM 
- |— Data nằm trong named volume 



Environment Variables & Secrets

```
dk@dk:~/docker-compose-lab$ docker compose config | grep -nE 'POSTGRES_(USER|PASSWORD|PASSWORD_FILE|DB)'\n7:     POSTGRES_DB: labdb\n8:     POSTGRES_PASSWORD_FILE: /run/secrets/db_password\n9:     POSTGRES_USER: labuser
```

```
dk@dk:~/docker-compose-lab$ docker inspect lab-db --format '{{range .Config.Env}}{{println .}}{{end}}' \n| grep -E 'POSTGRES_|PASSWORD'\nPOSTGRES_PASSWORD_FILE=/run/secrets/db_password\nPOSTGRES_DB=labdb\nPOSTGRES_USER=labuser
```